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CHARGING DEVICE AND SECURITY INSPECTION DEVICE

Abstract

The application disclosed a charging device and security inspection device. The body of the charging device is formed into a charging placing surface, and the first magnetic part is arranged on the side of the charging placing surface facing the inside of the charging device. The first magnetic suction portion is used to fix the metal detector onto the charging placement surface. When the metal detector needs to be picked up from the charging device, the metal detector can be picked up by grasping the handle of the metal detector overcoming the magnetic force of the first magnetic suction portion. Compared with traditional charging devices, there is no need to first take the metal detector's detection part and then convert the holding position.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to Chinese Application No. 202420398852.X, entitled “CHARGING DEVICE AND SECURITY INSPECTION DEVICE” filed on Mar. 1, 2024, which is hereby incorporated by reference herein as if set forth in its entirety.

TECHNICAL FIELD

[0002] The application relates to the field of security inspection device technology, specifically a charging device and security inspection device.

BACKGROUND OF THE APPLICATION

[0003] Handheld metal detectors are commonly used tools in security inspection, such as, handheld metal detectors are used in airports, train stations, and other important places to prevent the carrying of dangerous metal items. However, one of the significant challenges these devices face is the method of charging.

[0004] Traditional charging methods generally require placing the handle of the handheld metal detector in a charging slot. The process of placing the handheld metal detector require precise alignment and secure placement, which can be time-consuming and inconvenient, especially in high-traffic situations where quick response times are crucial.

DESCRIPTION OF THE APPLICATION

[0005] To solve the problems in the background, the application provides a charging device and a security inspection device, which feature an integrated magnetic design. This design make it easier and faster to pick up or place the handheld metal detector from the charging device.

[0006] The first objective of the application is to provide a charging device, the charging device comprises a body and a first magnetic suction portion; [0007] the body, formed with a charging placement surface, the charging placement surface is used to cooperate with a detection portion of a metal detector; [0008] the first magnetic suction portion, disposed on the body, and located on a side of the charging placement surface facing an inside of the charging device, the first magnetic suction portion is used to fix the metal detector onto the charging placement surface.

[0009] In one embodiment, the body further forms an avoiding surface, the avoiding surface connects to the charging placement surface, and the avoiding surface is tilted relative to the charging placement surface, and the avoiding surface deviates from the metal detector, the avoiding surface is used to avoid the detection portion of the metal detector.

[0010] In another embodiment, the charging placement surface is tilted relative to a bottom surface of the body, wherein an angle between the charging placement surface and the bottom surface of the body is 40°-50°.

[0011] In another embodiment, the body forms an storage cavity, wherein the storage cavity forms a positioning slot for fixing a sensitivity calibration tester.

[0012] In another embodiment, the body comprises an outer shell, an inner shell and a base, wherein the inner shell is disposed on the base and forms the storage cavity with the base, and the outer shell is sleeved outside the inner shell, [0013] the outer shell forms the charging placement surface, wherein a side of the charging placement surface facing the inner shell bas a mounting groove, and the first magnetic suction portion is embedded in the mounting groove; [0014] the inner shell has an installation surface corresponding to the charging placement surface, wherein the installation surface is provided with a charging portion; [0015] the base forms the positioning slot.

[0016] In another embodiment, a counterweight portion is disposed on the base; [0017] a counterweight tank is formed in the base, the counterweight portion is embedded in the counterweight tank, and the detachable cover of the counterweight tank is provided with a cover plate.

[0018] In another embodiment, the base forms a connecting groove on a side away from the inner shell, wherein an edge of the side away from the inner shell forms a connecting opening, and the connecting groove is connected to the connecting opening.
[0019] The second objective of application is to provide a security inspection device, the security inspection device comprises a metal detector and any one of the aforesaid the charging device;
[0020] the metal detector, having a detection portion and a handle portion, wherein the detection portion is provided with a second magnetic suction portion and an metal sensing coil, the second magnetic suction portion is used to cooperate with the first magnetic suction portion of the charging device, and the metal sensing coil is used to detect metal.
[0021] In one embodiment, the second magnetic suction portion is located at an end of the detection portion close to the handle portion, and the metal sensing coil is located at an end of the detection portion away from the handle portion.
[0022] In another embodiment, when the body of the charging device comprises an outer shell, an inner shell and a base, and a side of the base away from the inner shell forms a connecting groove, an edge of the side away from the inner shell forms a connecting opening, and the connecting groove is connected to the connecting opening; [0023] the security inspection device comprises at least two charging devices and at least one connecting member, wherein two ends of the connecting member are formed with avoiding slots, wherein the avoiding slots divide protruding ends on the connecting member, the protruding ends are embedded into the connecting grooves, and the edges of the base where the connecting openings are located are embedded into the avoiding slots.
[0024] The beneficial effects of the application are:
[0025] 1. When the metal detector needs to be picked up from the charging device, the metal detector can be picked up by grasping the handle of the metal detector overcoming the magnetic force of the first magnetic suction portion. Compared with traditional charging devices, there is no need to first take the metal detector's detection part and then convert the holding position.
[0026] 2. The body of the charging device forms a storage cavity, and the positing slot in the storage cavity can be used to place the sensitivity calibration tester, which is convenient for users to use the sensitivity calibration tester to calibrate the working state of the metal detector before using.
[0027] 3. A counterweight portion is arranged on the base of the charging device body. When the charging device is placed on the ground, the counterweight portion can ensure the stability of the charging device When charging device is fixed on the wall, the counterweight portion can be removed to reduce the weight of the charging device. This design improve the flexibility of charging device installation and use.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0028] To further clarify the implementation of this present application, a brief introduction to the drawings needed for describing the embodiments or existing technology follows. Obviously, the drawings described below represent only some embodiments of this present application. For those skilled in the art, other drawings can be obtained without creative effort based on these drawings.
[0029] FIG. **1** is a structure diagram of the first view of the charging device provided by an embodiment of the present application.
[0030] FIG. **2** is a structure diagram of the second view of the charging device provided by an embodiment of the present application.
[0031] FIG. **3** is an exploded view of the charging device provided by an embodiment of the present application.
[0032] FIG. **4** is a structure diagram of the housing of the charging device provided by an embodiment of the present application.

[0033] FIG. **5** is a structure diagram of one side of the base away from the inner shell of the charging device provided by an embodiment of the present application.
[0034] FIG. **6** is a structure diagram of the security inspection device provided by an embodiment of the present application.
[0035] FIG. **7** is an exploded view of the metal detector of the security inspection device by an embodiment of the present application.
[0036] FIG. **8** is a structure diagram of a plurality of charging devices connected by a connecting member of the security inspection device by an embodiment of the present application.
[0037] FIG. **9** is a structure diagram of the connecting member of the security inspection device by an embodiment of the present application.
LIST OF REFERENCES AND FIGURES
[0038] **100**—charging device; **10**—body; **11**—housing; **111**—charging placement surface; **111***a*—first side; **111***b*—second side; **112**—avoiding surface; **113**—mounting groove; **114**—limit protrusion; **12**—inner shell; **121**—installation surface; **13**—base; **131**—counterweight portion; **132**—counterweight tank; **133**—cover plate; **134**—positioning slot; **135**—connecting groove; **136**—connecting opening; **14**—storage cavity; **15**—charging portion; **20**—first magnetic suction portion; **200**—metal detector; **201**—detection portion; **202**—handle portion; **203**—second magnetic suction portion; **204**—metal sensing coil; **300**—connecting member; **301**—avoiding slot; **302**—protruding ends; **400**—sensitivity calibration tester.
PREFERRED EMBODIMENT OF THE APPLICATION
[0039] The technical solutions in the embodiments of the application will be clearly and completely described below with reference to the accompanying drawings. Apparently, the described embodiments are only some embodiments of the application, not all embodiments. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the application without creative efforts shall fall within the protection scope of the application.
[0040] The term “embodiment” mentioned herein means that the specific features, structures or characteristics described in combination with the embodiments may be included in at least one embodiment of the present application. The appearances of this phrase at various places in the specification do not necessarily all refer to the same embodiment, nor are they mutually exclusive, independent or alternative embodiments. Those skilled in the art understand explicitly and implicitly that the embodiments described herein may be combined with other embodiments.
[0041] Referring to FIGS. **1-3** and FIG. **6**, the present application provides a charging device **100** that can be used to charge a metal detector **200**. The charging device **100** comprises a body **10** and a first magnetic suction portion **20**. The body **10** of the charging device **100** is formed with a charging placement surface **111** for cooperating with the detecting portion **201** of the metal detector **200**. The first magnetic suction portion **20** is disposed on the body **10** and located on a side of the charging placement surface **111** facing the inside of the charging device **100**. The first magnetic suction part **20** is used to fix the metal detector **200** to the charging placement surface **111**.
[0042] It should be noted that the metal detector **200** generally includes a detecting portion **201** and a handle portion **202**, where the detecting portion **201** is provided with a metal sensing coil **204**. In use, the user holds the handle portion **202** of the metal detector **200** and scans the object to be inspected with the detecting portion **201** of the metal detector **200**. The placement surface formed by the charging device **100** provided in the present application is used to cooperate with the detecting portion **201** of the metal detector **200**. That is, the detecting portion **201** of the metal detector **200** cooperates with the charging placement surface **111**. The user's operation of placing and removing the metal detector **200** can be completely accomplished by holding the handle portion **202** of the metal detector **200** without switching the holding part of the metal detector **200**. It can be understood that the detecting portion **201** of the metal detector **200** has a second magnetic suction portion **203** cooperating with the first magnetic suction portion **20**. The first magnetic suction portion **20** and the second magnetic suction portion **203** may both be magnets, or one may

be a magnet and the other may be a metal that can be magnetically sucked. By providing the first magnetic suction portion **20** in the charging device **100** and using the cooperation between the first magnetic suction portion **20** of the charging device **100** and the second magnetic suction portion **203** of the metal detector **200**, not only can the metal detector **200** be fixed on the charging placement surface **111**, but also the placing position of the metal detector **200** on the charging placement Surface **111** can be corrected by magnetic force to ensure that the metal detector **200** is aligned with the charging portion **15** of the body **10**.

[0043] Since the detecting portion **201** of the general metal detector **200** has a certain size in the longitudinal direction, in the longitudinal direction of the metal detector **200**, a part of the detecting portion **201** can be in contact with the charging placement surface **111**, and the end of the detecting portion **201** away from the handle portion **202** protrudes from the charging placement surface **111**.

[0044] As shown in FIGS. **1** and **3**, in an embodiment, the body **10** also forms an avoiding surface **112**. The avoiding surface **112** connects to the charging placement surface **111**. The avoiding surface **112** is tilted relative to the charging placement surface **111**, and the avoiding surface **112** deviates from the metal detector **200**. The avoiding surface **112** is used to avoid the detection portion **201** of the metal detector **200**.

[0045] When the size of the detecting portion **201** of the metal detector **200** is relatively large compared to the size of the placement surface of the charging device **100** in its longitudinal direction, the end of the detecting portion **201** away from the handle portion **202** will protrude from the charging placement surface **111**. The avoiding surface **112** of the charging device **100** is used to avoid the detecting portion **201** of the metal detector **200**. For example, the charging placement surface **111** has opposite first side **111***a* and second side **111***b*. The metal detector **200** is placed on the charging placement surface **111** along the direction from the first side **111***a* to the second side **111***b*. The handle portion **202** of the metal detector **200** is located on the first side **111***a* of the charging placement surface **111**, and the end of the detecting portion **201** of the metal detector **200** away from the handle portion **202** protrudes from the second side **111***b* of the charging placement surface **111**. Correspondingly, the avoiding surface **112** is connected to the second side **111***b* of the charging placement surface **111**. As shown in FIGS. **1** and **3**, when the charging device **100** is in the normal use state, the avoiding surface **112** is tilted relative to the charging placement surface **111**, and the avoiding surface **112** deviates from the metal detector **200**, that is, the avoiding surface **112** is inclined downward to ensure the stability of the metal detector **200** placed on the charging placement surface **111**.

[0046] As shown in FIGS. **1** and **6**, in an embodiment, the charging placement surface **111** of the charging device **100** is inclined relative to the bottom surface of the body **10**, and the angle between the charging placement surface **111** and the bottom surface of the body **10** is 40°-50°. For example, the angle between the charging placement surface **111** and the bottom surface of the body **10** may be 40°, 45° or 50°, but is not limited thereto. By making the charging placement surface **111** inclined about 45 degrees relative to the bottom surface of the body **10**, it is convenient for the user to place and remove the metal detector **200**.

[0047] It should be noted that the charging placement surface **111** is inclined relative to the bottom surface of the body **10**, which means that the charging placement surface **111** is inclined along the direction from the first side **111***a* to the second side **111***b*. The first side **111***a* higher than the second side **111***b*, or the second side **111***b* higher than the first side **111***a*. The metal sensing coil **204** in the metal detector **200** is generally disposed at one end of the detecting portion **201** away from the handle portion **202**. In order to avoid the influence of the second magnetic suction portion **203** on the metal sensing coil **204**, the second magnetic suction portion **203** may be disposed at one end of the detecting portion **201** close to the handle portion **202**. Thus, in an embodiment, the second side **111***b* of the charging placement surface **111** is higher than the first side **111***a* to prevent the metal detector **200** from slipping off the first side **111***a* of the charging placement surface **111**. A limiting protrusion **114** can also be arranged on the first side **111***a* of the charging placement surface **111**.

[0048] As shown in FIGS. **1** and **2**, in an embodiment, the body **10** of the charging device **100** also forms an storage cavity **14** in which a positioning slot **134** is formed for fixing a sensitivity calibration tester **400**.
[0049] The body **10** of the charging device **100** is a shell structure with an storage cavity **14**, and the storage cavity **14** can be used for storage. It should be noted that in order to ensure the normal working status of the metal detector **200**, the metal detector **200** generally needs to use a sensitivity calibration tester **400** before daily use to calibrate the sensitivity of the metal detector **200**. By forming the positioning slots **134** in the accommodating cavity **14**, the sensitivity calibration tester **400** can be inserted into the positioning slot **134**. The positioning slot **134** may be a cylindrical groove, but is not limited thereto. The present application does not limit the specific number of positioning slots **134** in the accommodating cavity **14**, for example, two positioning slots **134** may be provided, corresponding to accommodating two sensitivity calibration testers **400**.
[0050] As shown in FIG. **3**, in an embodiment, the body **10** of the charging device **100** may include an outer shell **11**, an inner shell **12** and a base **13**. The inner shell **12** is disposed on the base **13** and forms the storage cavity **14** with the base **13**. The outer shell **11** is sleeved outside the inner shell **12**. The charging placement surface **111** of the body **10** is formed on the outer shell **11**. Referring also to FIG. **4**, the charging placement surface **111** of the outer shell **11** facing the inner shell **12** has a mounting groove **113**, and the first magnetic suction portion **20** of the charging device **100** is embedded in the mounting groove **113**. The inner shell **12** has an installation surface **121** corresponding to the charging placement surface **111** of the outer shell **11**. The installation surface **121** is provided with a charging portion **15**. The positioning slot **134** in the accommodating cavity **14** is formed in the base **13**. The positioning slot **134** on the base **13** is located in the accommodating cavity **14** after the inner shell **12** cooperates with the base **13**.
[0051] The installation surface **121** of the inner shell **12** corresponding to the charging placement surface **111** of the outer shell **11** means that after the outer shell **11** is sleeved outside the inner shell **12**, the installation surface **121** is directly below the charging placement surface **111**, so that the charging portion **15** on the mounting surface **121** can charge the metal detector **200** placed on the charging placement surface **111**. The charging portion **15** on the mounting surface **121** may charge the metal detector **200** on the charging placement surface **111** by electromagnetic induction, or may charge by contact between spring pins and the metal detector **200**, but is not limited thereto. It can be understood that the charging portion **15** also includes a charging control module for controlling the start and stop of the charging portion **15**. The charging control module may include an overheating protection circuit unit, a short circuit protection circuit unit, and an overcharge protection circuit unit.
[0052] The charging device **100** provided in this application can be placed on the ground or fixed on a vertical wall. In order to ensure the stability of the charging device **100** placed on the ground, a counterweight portion **131** is also arranged on the base **13**. In an embodiment, in order to realize the installation of the counterweight portion **131** on the base **13**, a counterweight tank **132** may be formed in the base **13**, and the counterweight portion **131** is embedded in the counterweight tank **132**. And the cover plate **133** is covered on the counterweight tank **132** detachably.
[0053] The counterweight portion **131** may be made of a material with a large density, for example, the material of the counterweight portion **131** may be iron, but is not limited thereto. By forming a counterweight groove **132** on the base **13**, placing the counterweight portion **131** in the counterweight groove **132**, and then covering the detachable cover plate **133** on the counterweight groove **132**, the detachability of the counterweight portion **131** is realized. When the charging device **100** needs to be installed on a vertical wall, the counterweight portion **131** can be removed from the counterweight groove **132**. The detachable mode of the cover plate **133** may be a snap-fit connection or a threaded fastener connection, but is not limited thereto. For easy installation and disassembly of the counterweight portion **131**, the detachable connection between the inner shell **12** and the base **13** and between the outer shell **11** and the base **13** may also be snap-fit or threaded

fasteners.
[0054] Referring also to FIG. **5**, in an embodiment, the base **13** forms a connecting groove **135** on a side away from the inner shell **12**, and an edge of the side away from the inner shell **12** forms a connecting opening **136**, and the connecting groove **135** is connected to the connecting opening **136**.
[0055] The base **13** of the body **10** is provided with a connecting groove **135** and a connecting opening **136** so that when multiple charging devices **100** are juxtaposed, the connecting member **300** can extend from the connecting opening **136** of the base **13** into the bottom of the base **13** to cooperate with the connecting groove **135**, thereby connecting adjacent charging devices **100** to improve stability. It should be noted that when multiple charging devices **100** are arranged side by side, both sides of the charging device **100** need to be connected to the adjacent charging device **100** through the connecting member **300**. It can be understood that the connecting groove **135** and the connecting opening **136** of the base **13** are formed on opposite sides of the base **13**.
[0056] Referring also to FIGS. **6** and **7**, in other embodiments, the present application also provides a security inspection device. The security inspection device includes a metal detector **200** and a charging device **100** in any of the foregoing embodiments. The metal detector **200** has a detecting portion **201** and a handle portion **202**. The detecting portion **201** is provided with a second magnetic suction portion **203** and an metal sensing coil **204**. The second magnetic suction portion **203** is used to cooperate with the first magnetic suction portion **20** of the charging device **100**. The metal sensing coil **204** is used to detect metal.
[0057] The second magnetic suction portion **203** in the detecting portion **201** is a magnet or metal that can be magnetically sucked. In order to avoid the influence of the second magnetic suction portion **203** on the metal sensing coil **204**, the second magnetic suction portion **203** is disposed at an end of the detecting portion **201** close to the handle, and the metal sensing coil **204** is disposed at an end of detecting portion **201** away from the handle.
[0058] Referring also to FIGS. **8** and **9**, in an embodiment, when the body **10** of the charging device **100** includes an outer shell **11**, an inner shell **12** and a base **13**, and a side of the base **13** away from the inner shell **12** forms The connecting tank **135** has a connecting opening **136** on the edge of the side away from the inner shell **12**, and the connecting groove **135** is connected to the connecting opening **136**. The security inspection device may include at least two charging devices **100** and at least one connecting member **300**. Both ends of the connecting member **300** are formed with avoiding grooves **301**. The avoiding grooves **301** divide protruding ends **302** on the connecting member **300**. The protruding ends **302** are embedded in the connecting grooves **135** of the base **13**, and the edges of the base **13** where the connecting openings **136** are located are embedded in the avoiding grooves **301**.
[0059] When the security inspection device has multiple charging devices **100**, each adjacent two charging devices **100** are connected by one connecting member **300**, which can connect multiple charging devices **100** to ensure stability of multiple charging devices **100** placed in the security inspection device.
[0060] The security inspection device provided in this application adopts all the technical solutions of all the embodiments of the charging device **100**, and thus at least has all the beneficial effects brought by the technical solutions of the embodiments of the charging device **100**, which will not be repeated here.

Claims

1. A charging device, comprises a body and a first magnetic suction portion; the body, formed with a charging placement surface, the charging placement surface is used to cooperate with a detection portion of a metal detector; the first magnetic suction portion, disposed on the body, and located on a side of the charging placement surface facing an inside of the charging device, the first magnetic

suction portion is used to fix the metal detector onto the charging placement surface.
2. The charging device according to claim 1, the body further forms an avoiding surface, the avoiding surface connects to the charging placement surface, and the avoiding surface is tilted relative to the charging placement surface, and the avoiding surface deviates from the metal detector, the avoiding surface is used to avoid the detection portion of the metal detector.
3. The charging device according to claim 1, the charging placement surface is tilted relative to a bottom surface of the body, wherein an angle between the charging placement surface and the bottom surface of the body is 40°-50°.
4. The charging device according to claim 1, the body forms an storage cavity, wherein the storage cavity forms a positioning slot for fixing a sensitivity calibration tester.
5. The charging device according to claim 4, the body comprises an outer shell, an inner shell and a base, wherein the inner shell is disposed on the base and forms the storage cavity with the base, and the outer shell is sleeved outside the inner shell; the outer shell forms the charging placement surface, wherein a side of the charging placement surface facing the inner shell has a mounting groove, and the first magnetic suction portion is embedded in the mounting groove; the inner shell has an installation surface corresponding to the charging placement surface, wherein the installation surface is provided with a charging portion; the base forms the positioning slot.
6. The charging device according to claim 5, a counterweight portion is disposed on the base; a counterweight tank is formed in the base, the counterweight portion is embedded in the counterweight tank, and the detachable cover of the counterweight tank is provided with a cover plate.
7. The charging device according to claim 5, the base forms a connecting groove on a side away from the inner shell, wherein an edge of the side away from the inner shell forms a connecting opening, and the connecting groove is connected to the connecting opening.
8. A security inspection device, comprises a metal detector and a charging device according to claim 1; the metal detector, having a detection portion and a handle portion, wherein the detection portion is provided with a second magnetic suction portion and an metal sensing coil, the second magnetic suction portion is used to cooperate with the first magnetic suction portion of the charging device, and the metal sensing coil is used to detect metal.
9. The security inspection device according to claim 8, the body further forms an avoiding surface, the avoiding surface connects to the charging placement surface, and the avoiding surface is tilted relative to the charging placement surface, and the avoiding surface deviates from the metal detector, the avoiding surface is used to avoid the detection portion of the metal detector.
10. The security inspection device according to claim 8, the charging placement surface is tilted relative to a bottom surface of the body, wherein an angle between the charging placement surface and the bottom surface of the body is 40°-50°.
11. The security inspection device according to claim 8, the body forms an storage cavity, wherein the storage cavity forms a positioning slot for fixing a sensitivity calibration tester.
12. The security inspection device according to claim 11, the body comprises an outer shell, an inner shell and a base, wherein the inner shell is disposed on the base and forms the storage cavity with the base, and the outer shell is sleeved outside the inner shell; the outer shell forms the charging placement surface, wherein a side of the charging placement surface facing the inner shell has a mounting groove, and the first magnetic suction portion is embedded in the mounting groove; the inner shell has an installation surface corresponding to the charging placement surface, wherein the installation surface is provided with a charging portion; the base forms the positioning slot.
13. The security inspection device according to claim 12, a counterweight portion is disposed on the base; a counterweight tank is formed in the base, the counterweight portion is embedded in the counterweight tank, and the detachable cover of the counterweight tank is provided with a cover plate.
14. The security inspection device according to claim 12, the base forms a connecting groove on a

side away from the inner shell, wherein an edge of the side away from the inner shell forms a connecting opening, and the connecting groove is connected to the connecting opening.

15. The security inspection device according to claim 8, the second magnetic suction portion is located at an end of the detection portion close to the handle portion, and the metal sensing coil is located at an end of the detection portion away from the handle portion.

16. The security inspection device according to claim 12, the security inspection device comprises at least two charging devices and at least one connecting member, wherein two ends of the connecting member are formed with avoiding slots, wherein the avoiding slots divide protruding ends on the connecting member, the protruding ends are embedded into the connecting grooves, and the edges of the base where the connecting openings are located are embedded into the avoiding slots.